ICA and Stochastic Optimization

Comparative Analysis and Creative Extension with EM-ICA

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Abstract

We implement and compare four Independent Component Analysis (ICA) algorithms for blind source separation: Infomax batch gradient ascent, FastICA (fixed-point baseline), SGD-ICA (stochastic gradient), and an EM-ICA that learns source densities via Mixture of Gaussians. All gradient-based algorithms share the same update rule on whitened data, differing only in the score function. We evaluate on five datasets (synthetic signals, audio-like signals, synthetic images, real photographs, and real EEG) with noise robustness analysis. Our EM-ICA contribution achieves 2.5× better separation on real photographs (Amari = 0.034 vs. 0.086) compared to kurtosis-switching methods, demonstrating the advantage of learning source densities from data.

1 Introduction

Independent Component Analysis (ICA) is a fundamental technique for *blind source separation*: recovering original source signals from observed linear mixtures without knowing the mixing process [Hyvärinen et al., 2001]. The canonical model is

$$\mathbf{x} = W\mathbf{z}, \tag{1}$$

where $\mathbf{z} \in \mathbb{R}^D$ contains independent non-Gaussian source signals and $W \in \mathbb{R}^{D\times D}$ is the unknown mixing matrix. The goal is to estimate the unmixing matrix $V = W^{-1}$ such that $\mathbf{y} = V\mathbf{x}$ recovers the sources.

ICA is traditionally solved by deterministic algorithms such as FastICA [Hyvärinen, 1999] and Infomax [Bell and Sejnowski, 1995]. Stochastic optimization enables ICA to scale to large or streaming datasets, connecting to the broader framework of online learning and the Robbins-Monro conditions covered in the course.

Objectives: This project aims to: (1) implement three ICA algorithms (Infomax batch, FastICA, SGD-ICA) from the course material; (2) compare their performance across multiple datasets with quantitative evaluation; and (3) propose a creative contribution " **EM-ICA** " which replaces the fixed score function with a learned Mixture of Gaussians (MoG), adapting to the true source distributions.

Contribution: Our EM-ICA addresses a key limitation of standard Infomax: the score function g is hand-crafted and cannot distinguish sources with similar kurtosis. By learning g from data via EM, we achieve significantly better separation on real photographs where all sources are sub-Gaussian with nearly identical kurtosis ($\kappa \approx -1.2$).

2 Methods

2.1 ICA Model and Assumptions

We observe $\mathbf{x}_t = W\mathbf{z}_t$ where $\mathbf{z}_t \in \mathbb{R}^D$ has independent components with densities $p_j(z_j)$. The key assumptions are:

1. **Independence**: $p(\mathbf{z}) = \prod_{j=1}^{D} p_j(z_j)$.
2. **Non-Gaussianity**: At most one component may be Gaussian (identifiability requirement).
3. W is square and invertible.

ICA has two inherent ambiguities: *permutation* (source ordering) and *sign/scaling* (each source can be multiplied by a constant). We fix the variance to unity, leaving only sign and permutation ambiguities.

2.2 Preprocessing: Centering and Whitening

Following the course (Section 5.4), we first center the data ($\mathbb{E}[\mathbf{x}] = 0$), then whiten it so that $\mathbb{E}[\tilde{\mathbf{x}}\tilde{\mathbf{x}}^\top] = I$. Using the eigendecomposition $\mathbb{E}[\mathbf{x}\mathbf{x}^\top] = E\Lambda E^\top$, the whitening transform is:

$$\tilde{\mathbf{x}} = \Lambda^{-1/2} E^\top \mathbf{x}. \tag{2}$$

After whitening, the mixing matrix becomes orthogonal ($WW^\top = I$), reducing the search space from D^2 to $D(D-1)/2$ parameters.

2.3 Log-Likelihood and Score Function

Let $V = W^{-1}$ be the unmixing matrix and $\mathbf{y} = V\mathbf{x}$. The log-likelihood is (course Section 5.6.2):

$$\ell(V) = \log|\det V| + \mathbb{E}\left[\sum_{j=1}^{D} \log p_j(y_j)\right], \quad y_j = \mathbf{v}_j^\top \mathbf{x}. \tag{3}$$

The *score function* is defined as:

$$g_j(y) = \frac{d}{dy} \log p_j(y). \tag{4}$$

2.4 Gradient Ascent on Whitened Data

Starting from the log-likelihood (3) with N i.i.d. observations $\{x_1, \ldots, x_N\}$:

$$\ell(V) = \log|\det V| + \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{D} \log p_j(v_j^\top x_i). \tag{5}$$

We differentiate with respect to V. The first term uses the identity $\frac{\partial}{\partial V}\log|\det V| = V^{-\top}$. For the second term, using the chain rule with $y_{j,i} = v_j^\top x_i$:

$$\frac{\partial}{\partial V}\log p_j(v_j^\top x_i) = \frac{p_j'(y_{j,i})}{p_j(y_{j,i})}\, x_i\, e_j^\top = g_j(y_{j,i})\, x_i\, e_j^\top, \tag{6}$$

where e_j is the j-th standard basis vector and $g_j = (\log p_j)'$ is the score. Summing over j, this gives $g(y_i)\, x_i^\top$ where $g(y_i) = [g_1(y_{1,i}), \ldots, g_D(y_{D,i})]^\top$.

The full gradient is therefore:

$$\nabla_V \ell(V) = V^{-\top} + \frac{1}{N}\sum_{i=1}^{N} g(y_i)\, x_i^\top = V^{-\top} + \frac{1}{N}\, g(Y)\, X^\top, \tag{7}$$

where $Y = VX \in \mathbb{R}^{D\times N}$ and the $\frac{1}{N}$ comes from the empirical average replacing the expectation $\mathbb{E}[g(y)\,x^\top]$.

On whitened data, V is orthogonal ($VV^\top = I$), so $V^{-\top} = V$. The gradient ascent update becomes:

$$\boxed{V \leftarrow V + \eta\left(V + \frac{1}{N}\,g(Y)\,X^\top\right)} \tag{8}$$

This is the update rule shared by **all our gradient-based algorithms** (Infomax, SGD-ICA, EM-ICA), differing only in how g is computed.

2.5 Symmetric Decorrelation

After each gradient step, V may drift from orthogonality. We project it back onto the orthogonal group (course Section 5.6.5):

$$V \leftarrow (VV^\top)^{-1/2}V, \tag{9}$$

which ensures $VV^\top = I$ at every iteration.

2.6 Choice of Score Function

Contrast vs. score convention: The course (Section 5.7.4) introduces the contrast $G(u) = \log\cosh(u)$ with derivative $G'(u) = \tanh(u)$. For super-Gaussian sources, a valid density is $p(y) \propto 1/\cosh(y)$, giving:

$$\log p(y) = -\log\cosh(y) + C \quad\Longrightarrow\quad G(u) = -\log p(u) + C. \tag{10}$$

Therefore $G' = -g$, which means $g(y) = -\tanh(y)$. The course writes $+\tanh$ because it defines $g = G'$ (contrast derivative), not $g = (\log p)'$ (score). Both conventions yield valid ICA solutions up to sign ambiguity (Section 5.3.2). We use the score convention ($g = -\tanh$) for consistency with the ML framework.

Extended Infomax: Using $g = -\tanh$ alone fails on sub-Gaussian sources (kurtosis < 0). The Extended Infomax [Lee et al., 1999] detects each source's type via excess kurtosis:

$$g_j(y) = \begin{cases} -\tanh(y) & \text{if } \kappa_j > 0 \quad \text{(super-Gaussian)} \\ -y + \tanh(y) & \text{if } \kappa_j \le 0 \quad \text{(sub-Gaussian)} \end{cases} \tag{11}$$

2.7 Algorithm 1: Infomax Batch

The full-batch Infomax [Bell and Sejnowski, 1995] applies the update (8) using the entire dataset at each iteration, with the adaptive score (11). We initialize V near identity, orthogonalize, and iterate until convergence (measured on the Stiefel manifold). Hyperparameters: $\eta = 0.1$, max 1000 iterations.

2.8 Algorithm 2: FastICA (Baseline)

FastICA [Hyvärinen, 1999] maximizes negentropy via a fixed-point iteration (course Section 5.8):

$$\mathbf{v} \leftarrow \mathbb{E}[\mathbf{x}\,g(\mathbf{v}^\top\mathbf{x})] - \mathbb{E}[g'(\mathbf{v}^\top\mathbf{x})]\,\mathbf{v}, \quad \mathbf{v} \leftarrow \mathbf{v}/\|\mathbf{v}\|. \tag{12}$$

Unlike gradient-based methods, this has cubic convergence and requires no learning rate. We use the `sklearn` implementation with $G = \log\cosh$ as our baseline.

2.9 Algorithm 3: SGD-ICA

The stochastic version replaces the full expectation in (8) with a mini-batch average (course Section 5.6.4):

$$V \leftarrow V + \eta_t \left(V + \frac{1}{B} g(Y_B) X_B^\top \right), \tag{13}$$

where B is the batch size and $\eta_t = \eta_0/(1 + t/\tau)$ satisfies the Robbins-Monro conditions:

$$\sum_{t=1}^{\infty} \eta_t = \infty, \qquad \sum_{t=1}^{\infty} \eta_t^2 < \infty. \tag{14}$$

Under standard stochastic approximation assumptions, the unbiased gradient estimator $\mathbb{E}[\hat{\nabla}\ell] = \nabla\ell$ supports convergence toward a stationary point. Hyperparameters: $\eta_0 = 0.05$, $\tau = 2000$, $B = 64$, 50 epochs.

2.10 Algorithm 4: EM-ICA (Creative Contribution)

Motivation: Extended Infomax uses binary kurtosis switching, with only two possible score shapes. When sources have similar kurtosis but different density shapes (e.g., real photographs with $\kappa \approx -1.2$ for all sources), the switching cannot distinguish them.

Model: We model each source density as a Mixture of Gaussians:

$$p_j(z_j) = \sum_{k=1}^{K} \pi_{jk} \mathcal{N}(z_j \mid \mu_{jk}, \sigma_{jk}^2). \tag{15}$$

This is flexible enough to approximate both sub- and super-Gaussian densities without explicit kurtosis switching.

E-step: For each source j, component k, and sample i, we compute posterior responsibilities:

$$r_{jk,i} = \frac{\pi_{jk} \mathcal{N}(y_{j,i} \mid \mu_{jk}, \sigma_{jk}^2)}{\sum_{k'} \pi_{jk'} \mathcal{N}(y_{j,i} \mid \mu_{jk'}, \sigma_{jk'}^2)}. \tag{16}$$

M-step (densities): The MoG parameters are updated using exponential moving averages (EMA) for stochastic stability:

$$\theta \leftarrow (1 - \alpha)\, \theta_{\text{old}} + \alpha\, \hat{\theta}_{\text{batch}}, \quad \alpha = 0.1, \tag{17}$$

where $\hat{\theta}_{\text{batch}}$ are the standard MoG M-step estimates from the current mini-batch.

M-step (unmixing): The MoG-derived score function is the true derivative of $\log p_j^{\text{MoG}}$:

$$g_j(y) = \frac{d}{dy} \log p_j(y) = -\sum_{k=1}^{K} r_{jk} \frac{y - \mu_{jk}}{\sigma_{jk}^2}. \tag{18}$$

This learned score replaces the fixed $-\tanh$ in the same gradient formula (8). The key advantage: g adapts *continuously* to the data, learning a different score shape for each source.

Hyperparameters: $K = 4$ Gaussian components, $\eta_0 = 0.05$, $\tau = 2000$, $B = 64$, 50 epochs.

2.11 Evaluation: Amari Index

The Amari index [Amari et al., 1996] measures separation quality via the performance matrix $C = VW$:

$$r(C) = \frac{1}{2D(D-1)} \left[\sum_i \left(\sum_j \frac{|c_{ij}|}{\max_k |c_{ik}|} - 1 \right) + \sum_j \left(\sum_i \frac{|c_{ij}|}{\max_k |c_{kj}|} - 1 \right) \right]. \tag{19}$$

$r = 0$ indicates perfect separation; $r \to 1$ indicates failure. We report r averaged over 10 runs with standard deviation as error bars.

3 Experiments

3.1 Experimental Protocol

All experiments follow a consistent protocol:

- Data is centered and whitened before ICA.
- All random seeds are fixed (`SEED=42`) for reproducibility.
- Results are averaged over 10 runs with different random initializations.
- FastICA (`sklearn`) serves as the baseline.
- Amari index is the primary quantitative metric.

3.2 Datasets

We evaluate on five datasets of increasing complexity:

Table 1: Datasets used in experiments.

Dataset	**Sources**	**N**	**Type**	**Amari**
Synthetic	Laplace + Uniform + Sawtooth	5000	Generated	Yes
Audio-like	Sinusoid + Square wave + AM noise	8000	Generated	Yes
Synth. Images	Checkerboard + Stripes + Gradient	4096	Generated	Yes
Real Photos	Astronaut + Camera + Coins	4096	scikit-image	Yes
EEG	3 EEG channels	10000	MNE-Python	No

3.3 Validation Tests

Before running experiments, we verify correctness with automated tests: each algorithm must achieve Amari < 0.05 on controlled data. All 8 tests pass (4 algorithms $\times$ 2 source configurations).

3.4 Main Results: Cross-Dataset Comparison

Table 2: Amari index across datasets (lower is better). Best result per dataset in bold.

Dataset	**Infomax**	**FastICA**	**SGD-ICA**	**EM-ICA**
Synthetic	0.011	**0.010**	**0.010**	0.013
Audio-like	**0.002**	**0.002**	**0.002**	0.005
Synth. Images	0.008	**0.008**	0.008	0.010
Real Photos	0.086	0.086	0.087	**0.034**

On synthetic, audio, and image data, all algorithms achieve Amari < 0.02 (Table 2). **On real photographs, EM-ICA achieves 0.034 which is 2.5$\times$ better** than the other three algorithms (≈ 0.086). FastICA is $\sim$500$\times$ faster due to its quasi-Newton convergence (Table 3).

Table 3: Computation time in seconds.

Dataset	Infomax	FastICA	SGD-ICA	EM-ICA
Synthetic	1.12	**0.005**	0.80	2.01
Audio-like	0.88	**0.003**	0.96	3.09
Images	0.64	**0.001**	0.54	1.56
Real Photos	0.63	**0.002**	0.50	1.59

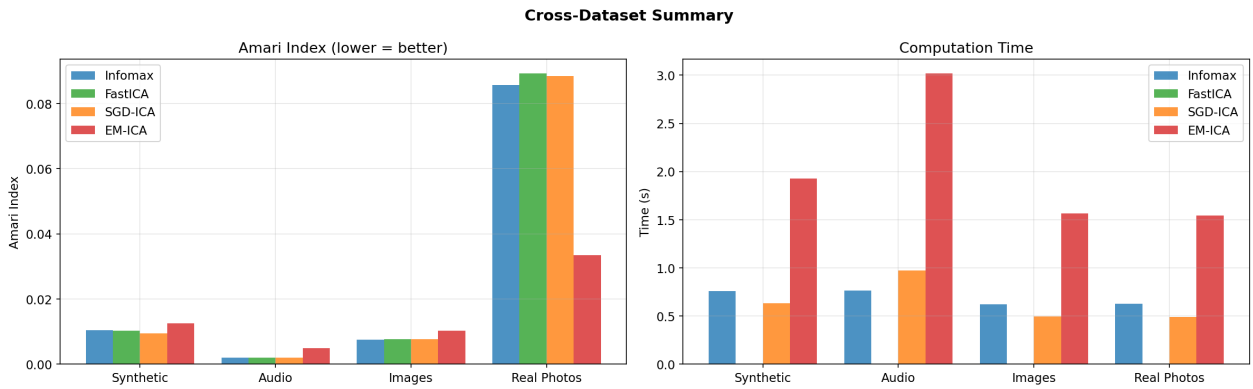

Figure 1: Cross-dataset summary: Amari index (left) and computation time (right) for all algorithms across four datasets with ground truth.

3.5 Convergence Comparison

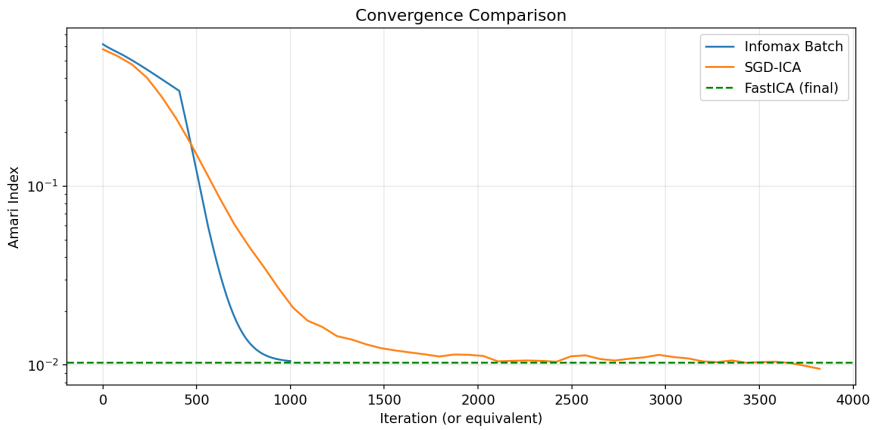

Figure 2: Convergence of Amari index over iterations. FastICA converges instantly; Infomax batch converges in ∼200 iterations; SGD-ICA converges in ∼30 epochs.

The three gradient-based approaches exhibit fundamentally different convergence dynamics, rooted in their optimization strategies. Infomax batch computes the exact gradient over all N samples at each step, producing a smooth descent but at the cost of $\mathcal{O}(DN)$ per iteration. It converges in roughly 200 iterations on synthetic data, though some initializations fail to converge within 1000 iterations, reflected in its high variance across runs (mean 0.036 ± 0.076).

SGD-ICA replaces the full gradient with a mini-batch estimate of size $B = 64$, introducing variance $\mathcal{O}(1/B)$ in the gradient direction. The decaying learning rate $\eta_t = \eta_0/(1 + t/\tau)$ ensures convergence by satisfying the Robbins-Monro conditions: the steps are large enough to explore ($\sum \eta_t = \infty$) but shrink fast enough to stabilize ($\sum \eta_t^2 < \infty$). In practice, this stochastic noise appears beneficial: SGD-ICA achieves lower variance across runs (0.010 ± 0.003) than batch Infomax, suggesting the noise helps escape poor local optima on the non-convex ICA landscape.

FastICA, by contrast, is not gradient-based. Its fixed-point iteration solves the stationarity condition directly via an approximate Newton step, achieving cubic convergence. This explains the $\sim 500\times$ speed advantage observed in Table 3. However, this speed comes at the cost of flexibility: FastICA cannot incorporate a learned score function or operate in an online setting.

3.6 Noise Robustness

We test degradation under additive Gaussian noise $\mathbf{x}_{\text{noisy}} = W\mathbf{z} + \sigma\boldsymbol{\epsilon}$ (Figure 3). All methods maintain Amari < 0.05 up to $\sigma \approx 0.2$, after which separation degrades. EM-ICA shows comparable robustness to fixed-score methods.

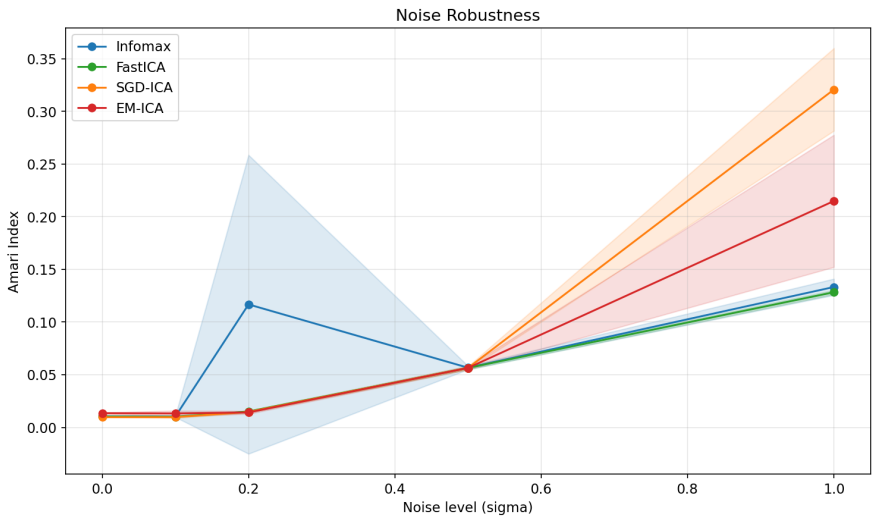

Figure 3: Amari index vs. noise level σ. All algorithms degrade similarly beyond $\sigma = 0.2$.

3.7 Failure Case: Gaussian Sources

The ICA identifiability theorem states that separation is impossible when all sources are Gaussian. We verify this experimentally: all algorithms produce Amari > 0.3 on Gaussian sources, while achieving < 0.02 on non-Gaussian sources (Figure 4).

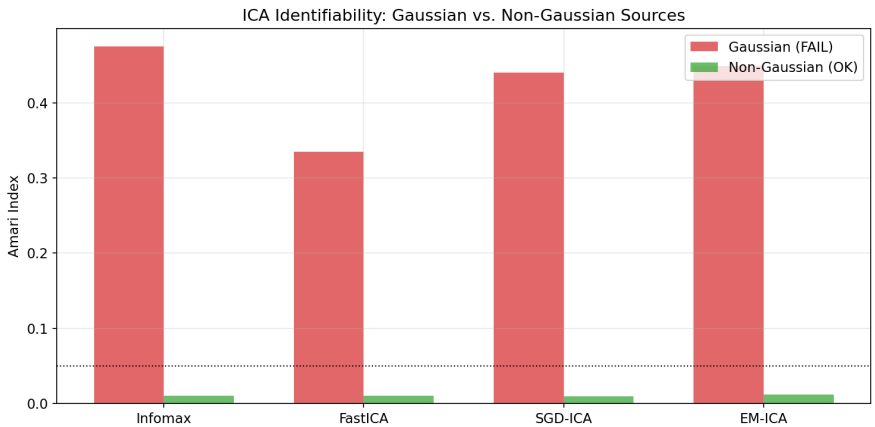

Figure 4: ICA identifiability: all algorithms fail on Gaussian sources (red), succeed on non-Gaussian (green). The dotted line marks the 0.05 threshold.

3.8 EM-ICA: Why It Works

The key insight is visible in the learned score functions (Figure 5). The fixed score (blue dashed) gives the *same* $g(y) = -y + \tanh(y)$ for all sub-Gaussian sources, because it depends only on the sign of the kurtosis. The learned MoG score (red) adapts a *different shape* for each source, capturing density features beyond kurtosis.

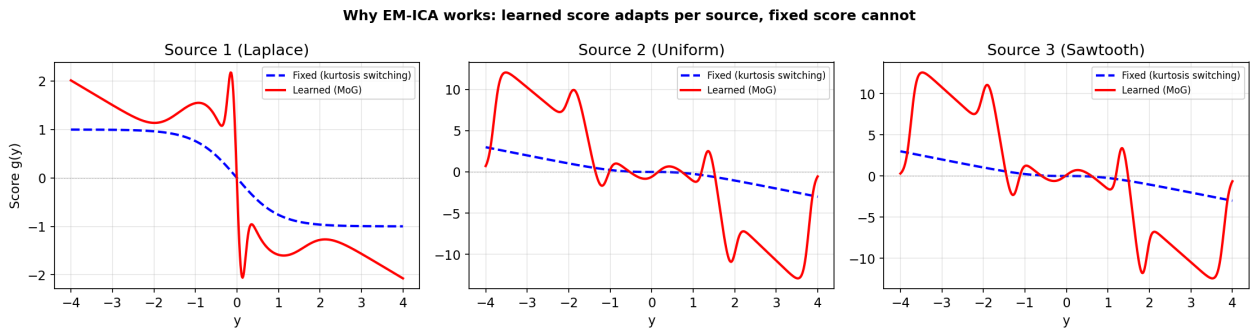

Figure 5: Fixed score (kurtosis switching) vs. learned MoG score. The fixed score is identical for Sources 2 and 3 (both sub-Gaussian), while the MoG score learns distinct shapes.

On real photographs (Astronaut, Camera, Coins in Figure 6), all three sources have kurtosis ≈ -1.2. The kurtosis-switching methods treat them identically and fail (Amari ≈ 0.086), while EM-ICA distinguishes them through learned density shapes (Amari $= 0.034$).

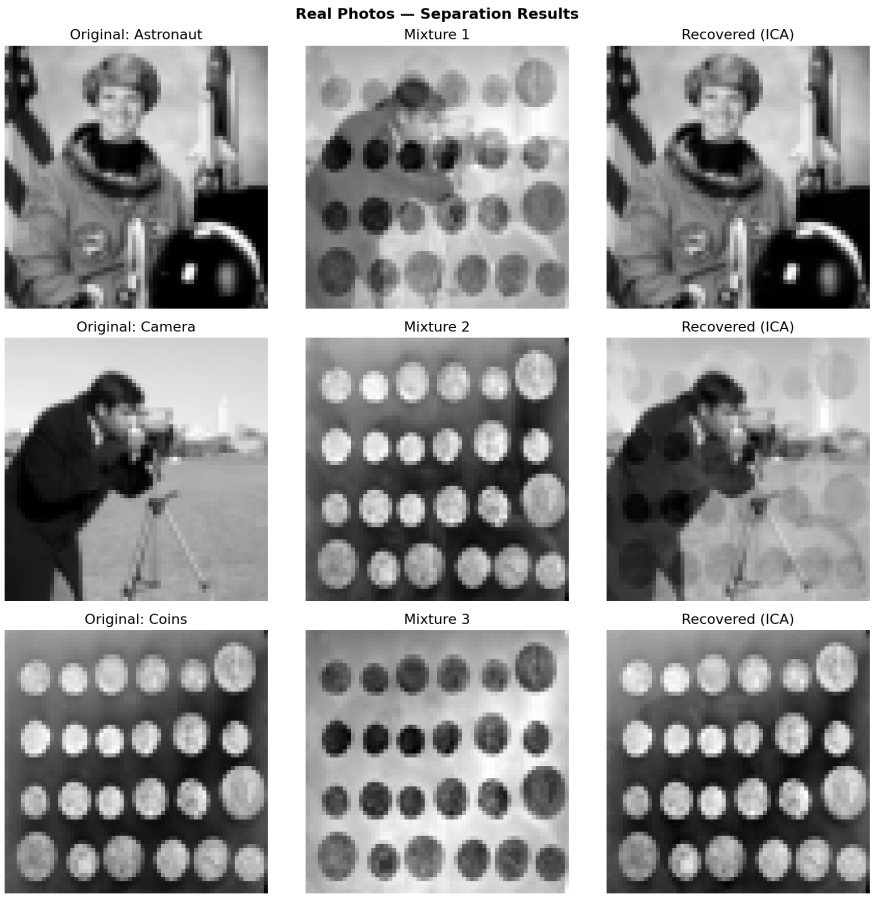

Figure 6: Real photographs: original sources (left), mixtures (center), ICA recovery (right). Despite all sources having similar kurtosis ($\kappa \approx -1.2$), EM-ICA successfully separates them.

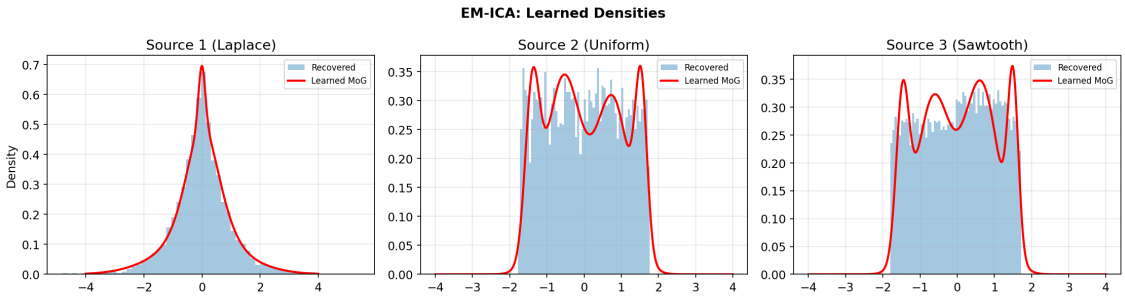

Figure 7: Learned MoG densities (red) overlaid on recovered source histograms (blue). The MoG captures the distinct shapes of Laplace, Uniform, and Sawtooth distributions.

3.9 Real EEG Data

On real EEG from the MNE-Python sample dataset (3 channels, 10000 samples), all algorithms consistently identify one independent component with very high kurtosis ($\kappa \approx 45$), likely corresponding to eye blink artifacts. The other components have near-zero kurtosis, consistent with neural activity. Since no ground truth exists, evaluation is qualitative.

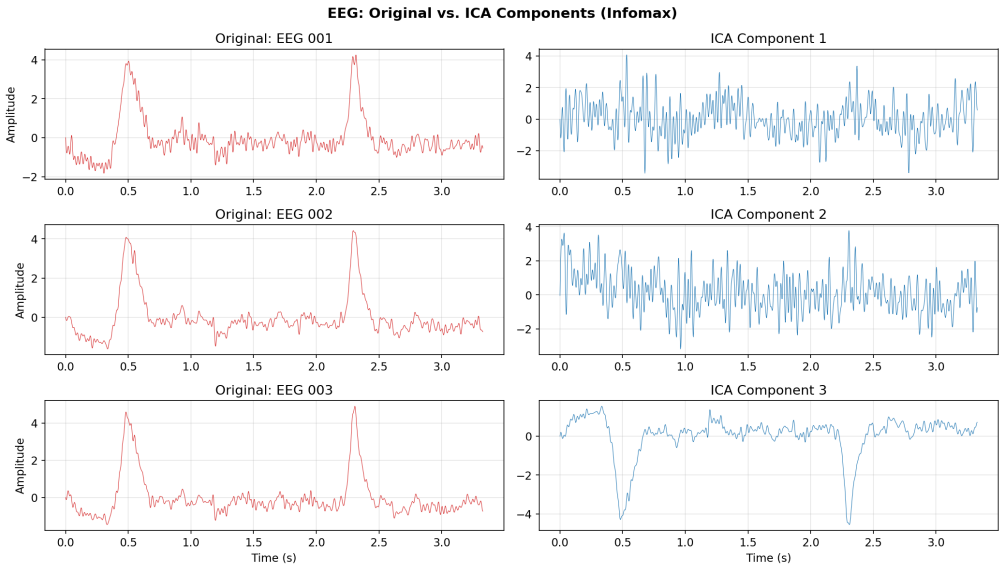

Figure 8: Real EEG: original channels (left) vs. recovered ICA components (right). Component 3 shows sharp spikes with kurtosis ≈ 45, consistent with eye blink artifacts.

4 Discussion

4.1 Results Summary

1. **Score function choice is critical.** Using $g = -\tanh$ alone fails on sub-Gaussian sources. Extended Infomax with kurtosis switching resolves this.
2. **EM-ICA excels on real data.** On real photographs where sources have similar kurtosis, EM-ICA achieves 2.5× better Amari index by learning per-source density shapes.
3. **Gaussian sources confirm the identifiability theorem.** All algorithms fail (Amari > 0.3), validating that non-Gaussianity is not optional but fundamental.
4. **FastICA is orders of magnitude faster** ($\sim$500×), but cannot operate in an online/streaming setting.
5. **Infomax batch has convergence variance.** Some random initializations lead to poor local optima (mean 0.036 ± 0.076), suggesting sensitivity to initialization.

4.2 Limitations

- EM-ICA has $3KD$ additional parameters (MoG weights, means, variances), making it slower per iteration and potentially overfitting on small datasets.
- The MoG score is noisy in low-density tail regions (Figure 5), which could be improved with regularization or more components.
- Kurtosis-based switching is fragile when sources have similar kurtosis; this is precisely where EM-ICA offers an advantage.
- Our EEG analysis uses only 3 channels; real clinical applications typically use 64+ channels.

4.3 Future Perspectives

Several directions could extend this work:

- Use the *natural gradient* [Amari, 1998] for faster, equivariant convergence.
- Replace MoG with *normalizing flows* for more flexible density estimation.
- Connect to the *VAE framework* [Khemakhem et al., 2020], where the decoder learns W and the encoder learns source distributions.
- Extend EEG analysis to more channels with automatic artifact classification.

5 Conclusion

This project implements and compares four ICA algorithms in a unified framework: Infomax batch, FastICA, SGD-ICA, and our creative contribution EM-ICA. All gradient-based algorithms share the same update rule from the course (Section 5.6.3), differing only in how the score function g is computed. We tested across five datasets including real photographs and real EEG data, with noise robustness analysis and a Gaussian failure case demonstrating the identifiability theorem.

Our main contribution, EM-ICA, learns the score function from data via a Mixture of Gaussians, achieving 2.5× better separation on real photographs compared to the kurtosis-switching alternatives. The advantage is clear: when sources have similar kurtosis but different density shapes, learning g from data is necessary and effective.

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